Solving by Graphing

Name _____

Algebra 2 Lesson 1-5 (fast track) Date _____ Period _____

Learning goal: I can solve an equation or inequality by graphing both sides and reading where the graphs meet.

Language goal: I can say "the solution is where the graphs intersect" and "the inequality is true where one graph is above (or below) the other."

On the quiz: You will (1) sketch two graphs to solve an equation and read the solution(s) off the intersection, (2) graph two cost equations and write the inequality for when one plan is cheaper, and (3) approximate where two functions are equal from a table of values.

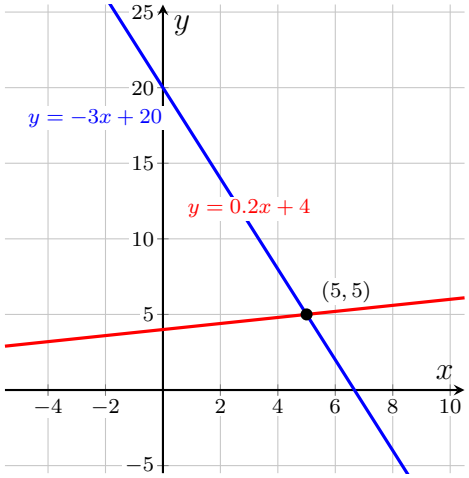
On the Topic 1 test: Question 11 asks you to solve an equation from a graph already drawn (a V and a parabola): write the x-value of every point where they cross (Example 1).

Example 1: Use a Graph to Solve an Equation

The idea: graph $y =$ left side and $y =$ right side. The x-value of each **intersection** is a solution.

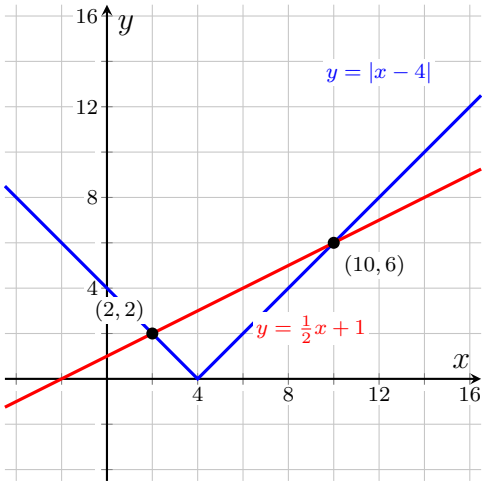
A. Solve $-3x + 20 = 0.2x + 4$ by graphing.

1. Graph $y = -3x + 20$ and $y = 0.2x + 4$.
2. Find the intersection: $(5, 5)$.
3. Solution: $x = 5$. Check: $-15 + 20 = 5$ and $1 + 4 = 5$. ✓



B. Solve $|x - 4| = \frac{1}{2}x + 1$ by graphing.

1. Graph $y = |x - 4|$ (a V, vertex $(4, 0)$) and $y = \frac{1}{2}x + 1$.
2. Two intersections: $(2, 2)$ and $(10, 6)$.
3. Solutions: $x = 2$ and $x = 10$. Check: $|2-4| = 2$ and $\frac{1}{2}(2) + 1 = 2$. ✓



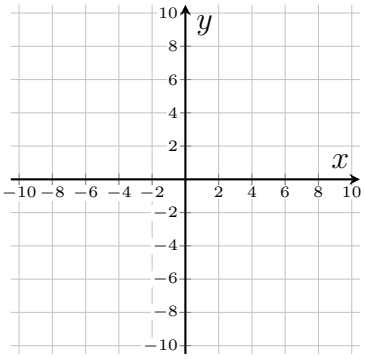
Say it: In Part B the solutions are $x =$ _____ and $x =$ _____ because . . .

Try It 1: Use a graph to solve each equation

Write the two $y =$ equations, graph both, mark each intersection with a dot, and write the solution(s).

a. $5x - 12 = 3$

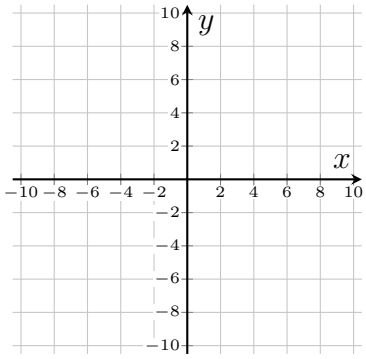
$y =$ _____ $y =$ _____



Solution: $x =$ _____

b. $-|x - 2| = -\frac{1}{2}x - 2$

$y =$ _____ $y =$ _____



Solution(s): $x =$ _____

How did you know where the solutions were? Use the word *intersect*.

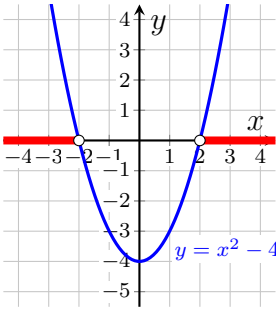
Example 2: Solve a One-Variable Inequality by Graphing

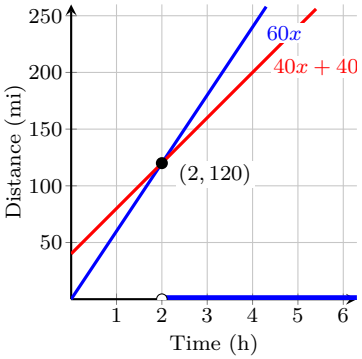
The idea: graph both sides. "Left > right" is true where the left graph is **above** the right one; intersections are the boundaries.

A. Solve $x^2 - 4 > 0$.

1. Graph $y = x^2 - 4$ and $y = 0$ (the x-axis).
2. Boundaries: the graph crosses the x-axis at $x = -2$ and $x = 2$.
3. The parabola is **above** the x-axis when $x < -2$ or $x > 2$ (thick part of the axis). Open circles: ± 2 give 0, not > 0.

Solution: $x < -2$ or $x > 2$.





B. A motorcycle is 40 mi ahead of a car. The car goes 60 mph, the motorcycle 40 mph. When is the car ahead?

1. Let $x =$ hours. Car: $60x$ mi. Motorcycle: $40x + 40$ mi (it *started* 40 mi ahead, so add 40).
2. Car ahead means $60x > 40x + 40$. Graph $y = 60x$ and $y = 40x + 40$. They intersect at $(2, 120)$; after that the $60x$ line is **above**.

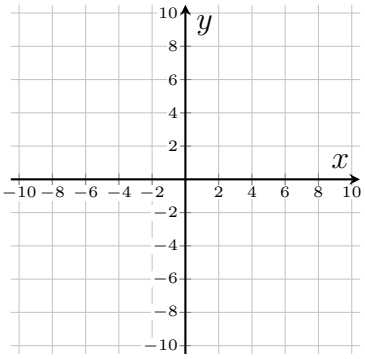
Solution: $x > 2$. The car is ahead any time after 2 hours.

Try It 2: Use a graph to solve each inequality

Graph both sides, dot the intersection(s), then mark the solution on the x-axis and write it.

a. $x^2 + 6x + 5 \geq 0$

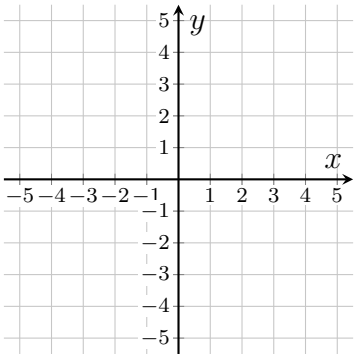
$y =$ _____ $y =$ _____



Solution: _____

b. $x + 3 > 7 - 3x$

$y =$ _____ $y =$ _____



Solution: _____

In part b, which graph has to be on top? Why?

Practice: Two cost plans

A gym charges \$12 per visit with no membership. With a \$60 membership, each visit costs only \$4.

a. Let x = number of visits. Total cost of each plan:
No membership: $y =$ _____
Membership: $y =$ _____

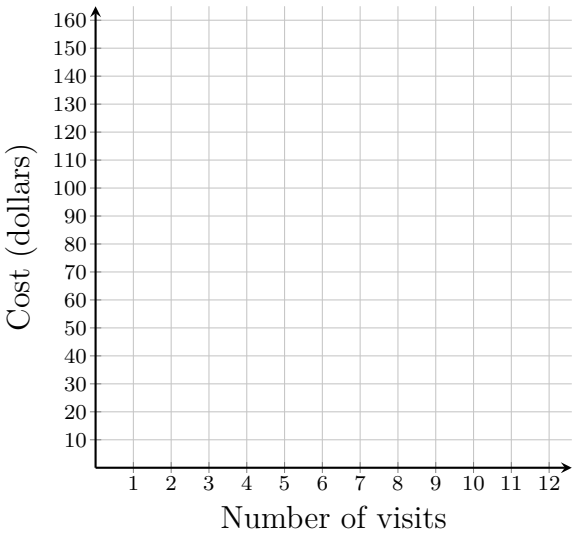
b. Graph both on the grid. Dot the intersection.

c. Inequality that says *the membership is cheaper*:

_____

d. Fewest visits for which the membership is cheaper: _____ visits

e. Where do the plans cost the same? Can you make that many visits?



IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred skill replaces your quiz score.**

Code	Skill	Feeds
M69 ★	Solve a system of equations by graphing	Quiz 1–3: graph both sides, read the intersection
FM2 ★	Approximate solutions using a table	Quiz 5: find where $f(x)$ and $g(x)$ are closest
ZBH	Find solutions using a table	Quiz 5 warm-up
39B	Graph solutions to absolute value equations	Try It with $\|x\|$ (optional)

Before the quiz, finish these sentences.

The solution to an equation is where the two graphs _____.

An inequality like $f(x) > g(x)$ is true where the graph of f is _____ the graph of g.

A membership is cheaper than pay-per-visit when its line is _____ the other line.